

$\sigma = 0.2$ · Noisy PSNR = 16.71 dB

SEEING THROUGH THE NOISE

How SVD Reveals Hidden Structure in Images

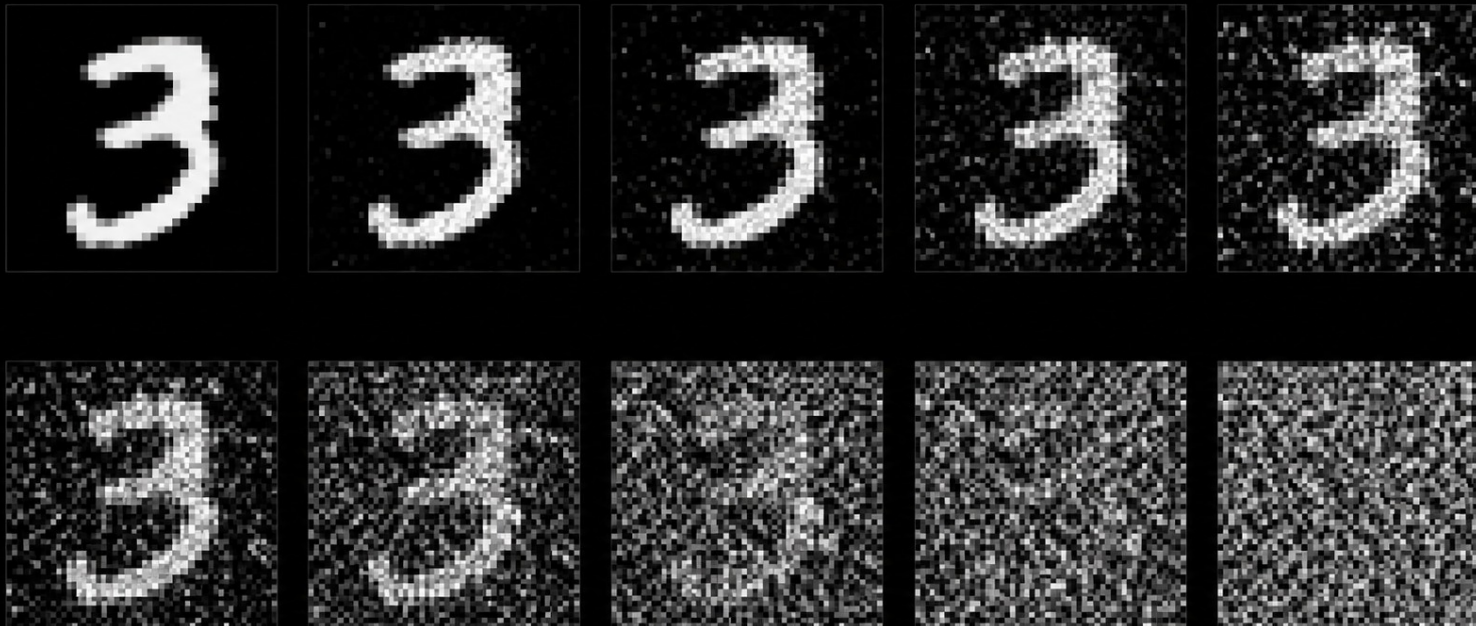
A story of mathematics, handwritten digits, and discovery.

What if you could look at a noise-corrupted image and mathematically recover the original digit?



CAN YOU READ THESE DIGITS?

They are all handwritten 3s... but noise makes them almost unreadable.



Notebook data

6,131

digit-3 images

28×28=784

pixels per image

$\sigma = 0.2$

Gaussian noise

16.71 dB

Noisy PSNR

Question for students:

Can you identify the number in each image?

SVD: THE TOOL THAT REVEALS STRUCTURE

It decomposes a matrix into "layers of information."

$$X = U \cdot \Sigma \cdot V^T$$

X

6131 × 784

All images

=

U

6131 × 6131

Image patterns

•

Σ

6131 × 784

Importance / energy

•

V^T

784 × 784

Pixel patterns

Analogy: imagine an orchestra.

U and V are the musicians; Σ is the volume of each instrument. The loudest instruments are the ones that matter most.

WHY DO THE DIMENSIONS MATTER?

The formula is simple; the dimensions tell the story.

Data matrix X

**6131 rows
= images**

**784 columns
= pixels**

$[6131 \times 784]$

$$X_k = U_k \cdot \Sigma_k \cdot V_k^T$$

U_k

$6131 \times k$

Σ_k

$k \times k$

V_k^T

$k \times 784$

$$[6131 \times 784] = [6131 \times k] \cdot [k \times k] \cdot [k \times 784]$$

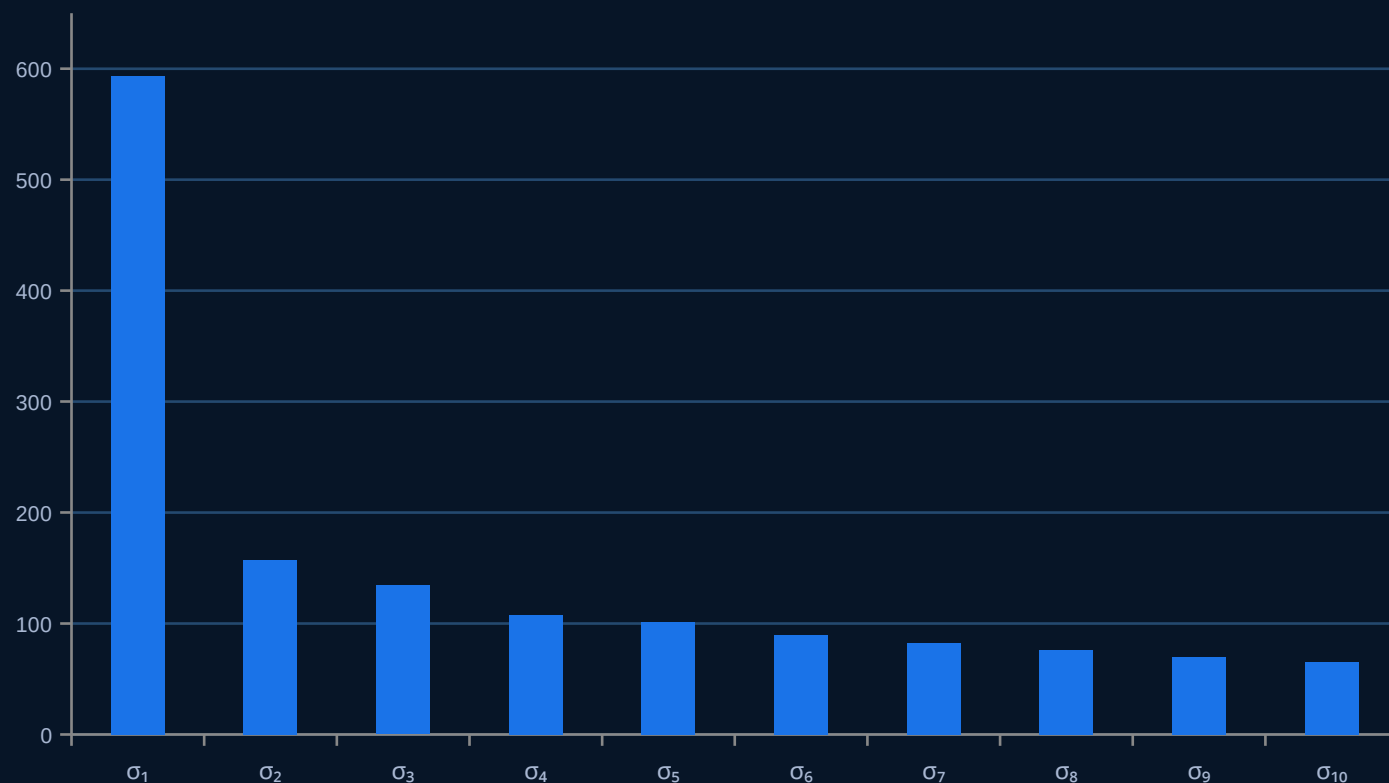
Choose k

keep only the k
most
important
patterns and
discard the
rest.

less data · same structure

THE SECRET: SEPARATING SIGNAL FROM NOISE

The digit shape lives in the large singular values. Noise is spread across many small values.



SIGNAL

first components
with the most energy

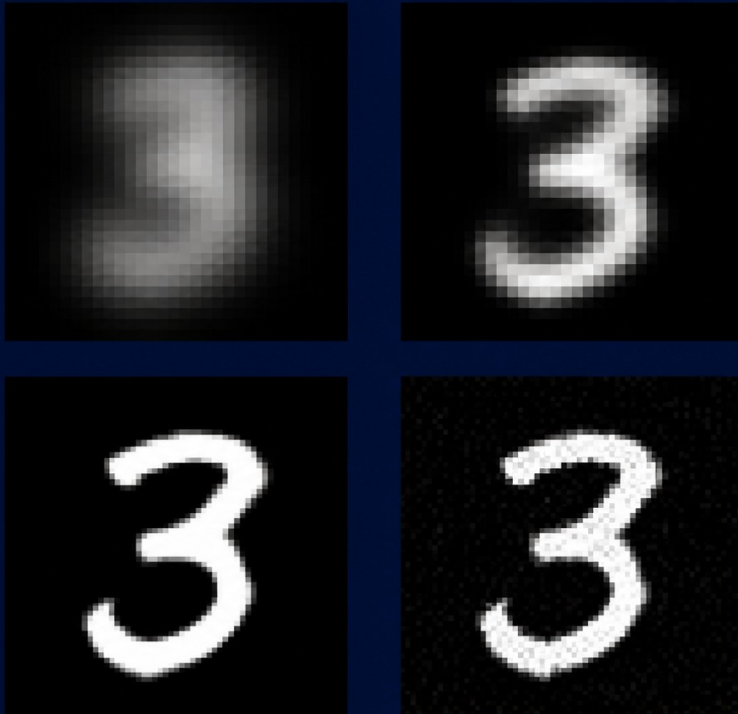
NOISE

many small,
dispersed components

$\sigma_1 = 593.72$ dominates the scale: that is why ranking matters.

HOW k TRANSFORMS THE IMAGE

4 reconstructions, 4 different outcomes



$k = 1$

PSNR 12.38 dB · Energy
57.2%
Almost unrecognizable

$k = 5$

PSNR 14.00 dB · Energy
67.6%
Starting to emerge

Visual takeaway: increasing k helps... until it starts bringing noise back.

$k = 20$

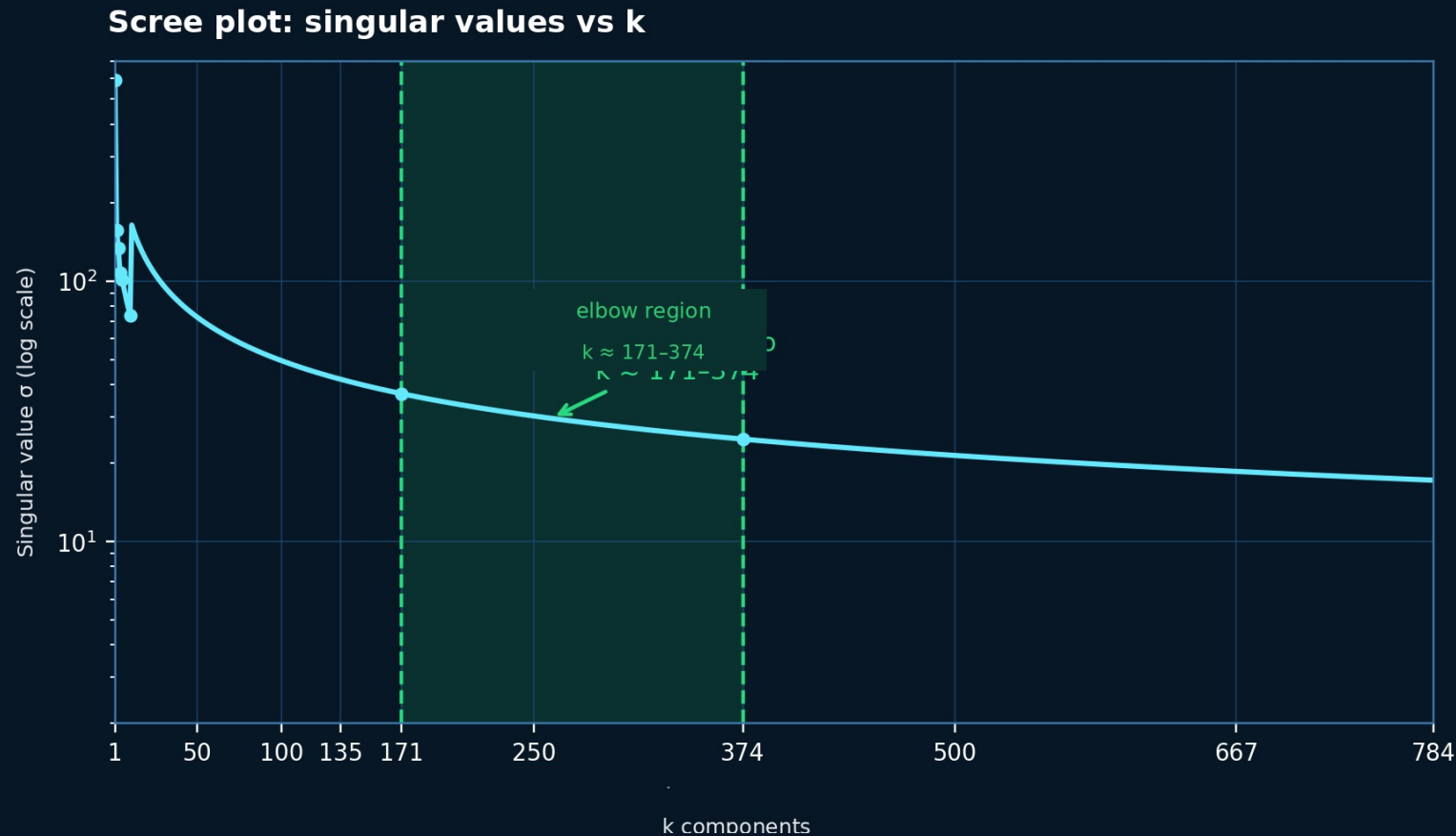
PSNR 16.23 dB · Energy
77.4%
Perfectly readable!

$k = 40$

PSNR 17.60 dB · Energy
81.9%
Noise returns

THE ELBOW OF THE SINGULAR VALUES

Method 1: find where the curve stops dropping rapidly.



Reading the scree plot

Values in descending order
We look for the curve's "elbow"
Suggested region: $k \approx 171-374$

k = 171

90% energy

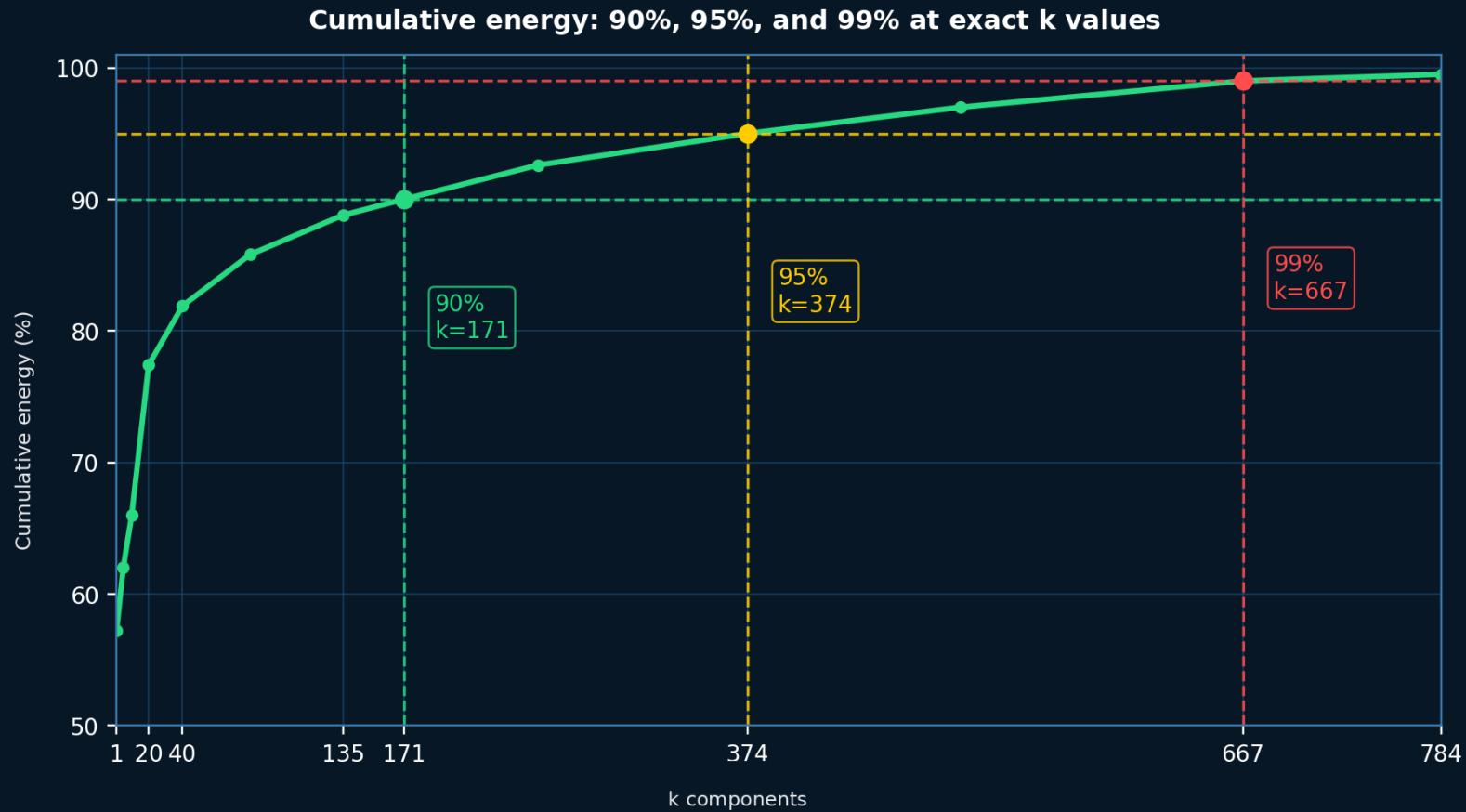
k = 374

95% energy

The elbow marks where values begin to appear less significant.

HOW MUCH ENERGY DOES EACH k CAPTURE?

Method 2: cumulative energy.



Key idea: the curve rises quickly at first and then levels off.

Results

90% → k=171

energy

95% → k=374

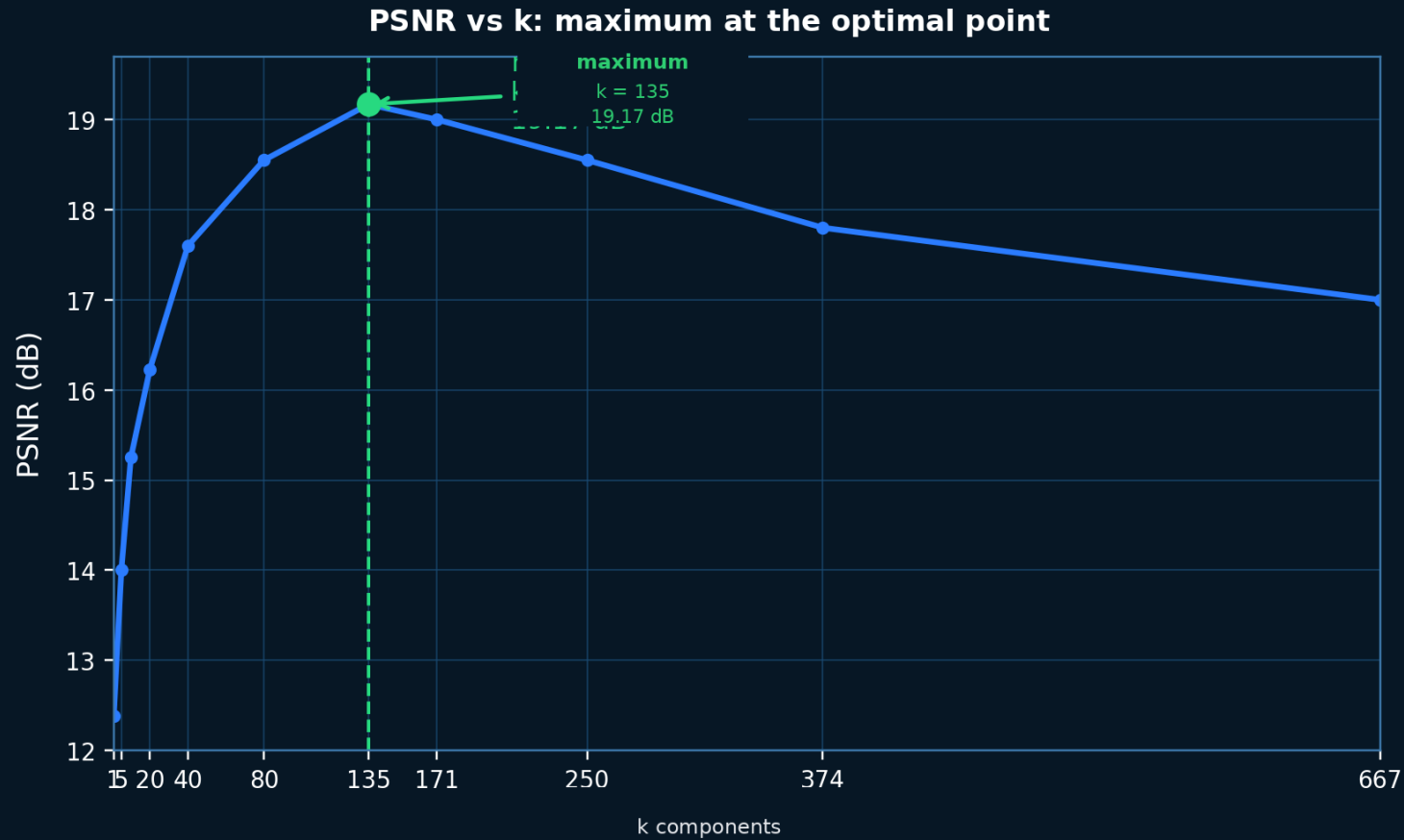
energy

99% → k=667

energy

THE PERFECT k: MAXIMUM QUALITY

Method 3: PSNR compares the reconstruction with the clean image.



optimal k = 135

Maximum PSNR = 19.17 dB

Improvement over noise: +2.46 dB

Comparison

95% energy	374
Elbow	171–374
PSNR	135

PSNR gives the most direct answer when we have a clean reference image.

NOT TOO LITTLE, NOT TOO MUCH: JUST THE RIGHT k !



small k

loses detail
blurry image

optimal k

clean image
perfectly readable

large k

noise reintroduced
looks grainy

Closing thought

"SVD does more than clean images: it teaches us that behind the chaos, structure is always waiting to be revealed."

Real-world applications

Neuroimaging: denoise scans
Satellites: enhance imagery
Film: restore movies
Facial recognition

Experiment: add noise and find your optimal k